Negative Binomial Distribution

Note that,

$$(1-q)^{-r} = \sum_{x=0}^{\infty} \binom{x+r-1}{x} q^x, \quad \text{for } |q| < 1, (\text{Taylor's Expansion}), r > 0, \text{real no}$$

This is called Binomial Theorem for negative index.

$\sum_{x=0}^{\infty} \binom{x+r-1}{x} p^r q^x = 1$ and the terms of the series forms a PMF. The probability distribution given by this PMF is known as Negative Binomial Distribution.

1 Definition

A counting random variable is said to follow a negative Binomial Distribution if its PMF is given by,

$$f(x) = \begin{cases} \binom{x+r-1}{x} p^r q^x, & x = 0, 1, 2 \dots \\ 0, & \text{otherwise} \end{cases}$$

2 Moments

Note that, $(1-q)^{-r} = \sum_{x=0}^{\infty} \binom{x+r-1}{x} q^x$

Differentiating wrt q,

$$(-r)(1-q)^{-r-1}(-1) = \sum_{x=0}^{\infty} \binom{x+r-1}{x} x q^{x-1}$$

$$\Rightarrow r p^{-r-1} = \sum_{x=1}^{\infty} \binom{x+r-1}{x} x q^{x-1}$$

Again, differentiating wrt q,

$$r(r+1)(1-q)^{-r-2} = \sum_{x=2}^{\infty} \binom{x+r-1}{x} x(x-1) q^{x-2}$$

$$\begin{aligned} E(X) &= \sum_{x=0}^{\infty} x \binom{x+r-1}{x} p^r q^x \\ &= p^r q \sum_{x=1}^{\infty} x \binom{x+r-1}{x} q^{x-1} \\ &= p^r q r p^{-r-1} \\ &= \frac{rq}{p} \end{aligned}$$

$$\begin{aligned} V(X) &= E\left(X^2\right) - E^2(X) \\ &= E(X(X-1)) + E(X) - E^2(X) \\ &= p^r q^2 \sum_{x=2}^{\infty} x(x-1) \binom{x+r-1}{x} q^{x-2} + \frac{rq}{p} - \frac{r^2 q^2}{p^2} \\ &= p^r q^2 r(r+1) p^{-r-2} + \frac{rq}{p} - \frac{r^2 q^2}{p^2} (\text{Derivative of sum = sum of derivatives, but it is not always true}) \end{aligned}$$

$$
\begin{aligned}
&= r(r+1)\frac{q^2}{p^2} + \frac{rq}{p} - \frac{r^2q^2}{p^2} \\
&= \frac{rq}{p}\left[\frac{rq}{p} + \frac{q}{p} + 1 - \frac{rq}{p}\right] \\
&= \frac{rq}{p^2} \\
&\therefore E(X) = \frac{rq}{p} < \frac{rq}{p^2} = \mathrm{Var}(X)
\end{aligned}
$$

3 Recursion Relation of Negative Binomial Distribution

$$
\begin{aligned}
\mu_m = E[X - E(X)]^m &= E\left[X - \frac{rq}{p}\right]^m \\
&= \sum_{x=0}^{\infty}\left(x - \frac{rq}{p}\right)^m \binom{x+r-1}{x} p^r q^x
\end{aligned}
$$

Differentiating wrt p we get,

$$
\begin{aligned}
\frac{\partial \mu_m}{\partial p} =& \sum_{x=0}^{\infty}\left(x - \frac{rq}{p}\right)^m \binom{x+r-1}{x} r p^{r-1} q^x + \sum_{x=0}^{\infty}\left(x - \frac{rq}{p}\right)^m (x+r-1) p^r x q^{x-1}(-1) \\
&+ \sum_{x=0}^{\infty} m\left(x - \frac{rq}{p}\right)^{m-1} \binom{x+r-1}{x} p^r q^x (-r)\frac{p(-1) - (1-p)}{p^2} \\
=& \sum_{x=0}^{\infty}\left(x - \frac{rq}{p}\right)^{m-1} \binom{x+r-1}{x} p^r q^x \left(\frac{mr}{p^2}\right) + \sum_{x=0}^{\infty}\left(x - \frac{rq}{p}\right)^m (x+r-1) p^r q^x \left[\frac{r}{p} - \frac{x}{a}\right] \\
=& \frac{mr}{p^2}\mu_{m-1} - \frac{1}{a}\mu_{m+1} \\
\Rightarrow \frac{1}{q}\mu_{m+1} &= \frac{mr}{p^2}\mu_{m-1} - \frac{\partial \mu_m}{\partial p} \\
\Rightarrow \mu_{m+1} &= q\left(\frac{mr}{p^2}\mu_{m-1} - \frac{\partial \mu_m}{\partial p}\right)
\end{aligned}
$$

This is the recursion relation of Negative Binomial distribution.
Now, $\mu_0 = 1, \quad \mu_1 = 0$
If $m = 1, \mu_2 = q\left(\frac{r}{p^2}\mu_0 - 0\right) = \frac{rq}{p^2}$

$$
\begin{aligned}
m = 2, \quad \mu_3 &= q\left(\frac{2r}{p^2}\mu_1 - \frac{\partial \mu_2}{\partial p}\right) \\
&= q\left(\frac{2r}{p^2}\cdot 0 - r\frac{(p-2)}{p^3}\right) \\
&= qr\frac{(2-p)}{p^3} \\
\therefore \text{ skewness } = \frac{\mu_3}{\mu_2^{3/2}} &= \frac{qr\frac{(2-p)}{p^3}}{\left(\frac{qr}{p^2}\right)^{3/2}} = \frac{1+q}{\sqrt{qr}} > 0
\end{aligned}
$$

$\Rightarrow$ The distribution is positively skewed.

$$
\begin{aligned}
\mu_4 &= q\left\{\frac{3r}{p^2}\mu_2 - \frac{2\mu_3}{2p}\right\} \\
&= q\left\{3q\left(\frac{r}{p^2}\right)^2 - \left\{-\frac{r}{p^2}\left(6p^{-2}-6p^{-1}+1\right)\right\}\right\} \\
&= 3\left(\frac{qr}{p^2}\right)^2 + \frac{rq}{p^2}\left(\frac{6}{p^2}-\frac{6}{p}+1\right) \\
&= \frac{rq\left(1+4q+q^2\right)+3r^2q^2}{p^4} \\
\therefore \text{ Kurtosis } = \frac{\mu_4}{\mu_2^2} - 3 &= \frac{rq\left(1+4q+q^2\right)+3r^2q^2}{\frac{q^2r^2}{p^4}p^4} - 3 \\
&= \frac{1+4q+q^2}{q^r} + 3 - 3 > 0
\end{aligned}
$$

4 Problems

1. If $X \sim \text{NB}(r,p)$, show that $E\left(X^k\right) = \frac{rq}{p}E(Y+1)^{k-1}$ where $Y \sim \text{NB}(r+1,p)$. Hence, find $E(X)$ and $\text{Var}(X)$.

$$
\begin{aligned}
E\left(X^k\right) &= \sum_{x=0}^{\infty} x^k \binom{x+r-1}{x} p^r q^x \\
&= \sum_{x=1}^{\infty} x^{k-1} \frac{(x+r-1)\ldots(r+1)r}{(x-1)!} p^r q^x \\
&= \frac{rq}{p}\sum_{x=1}^{\infty} x^{k-1} \frac{(\overline{x-1}+\overline{r+1}-1)\ldots(r+1)}{(x-1)!} p^{r+1} q^{x-1} \\
&= \frac{rq}{p}\sum_{x-1=y=0}^{\infty} (y+1)^{k-1} \frac{(y+\overline{r+1}-1)\ldots(r+1)}{y!} p^{r+1} q^{y} \\
&= \frac{rq}{p}\sum_{y=0}^{\infty} (y+1)^{k-1} \underbrace{\binom{(y+\overline{r+1}-1)}{y} p^{r+1} q^{y}} \\
&= \frac{rq}{p} E(Y+1)^{k-1}
\end{aligned}
$$

$$
\begin{aligned}
E(X) &= \frac{rq}{p}E(Y+1)^0 = \frac{rq}{p} \\
E\left(X^2\right) &= \frac{rq}{p}(E(Y)+1) = \frac{rq}{p}\left(\frac{(r+1)q}{p}+1\right) \\
&= \frac{rq}{p}\frac{rq+q+p}{p} = \frac{rq(rq+1)}{p^2}
\end{aligned}
$$

2. Derive the PMF of NB distribution in the following form.

$$
f(x) = \binom{-r}{x} P^x Q^{-r-x}, \quad x = 0,1,2,\ldots, P+Q=1
$$

Show that the mean and variance could be written as the corresponding formula of Binomial Distribution.

$$
\begin{aligned}
f(x) &= \binom{x+r-1}{x} p^r q^x \\
&= \frac{(x+r-1)\cdots(r+1)r}{x!}\left(\frac{1}{p}\right)^{-r} q^x
\end{aligned}
$$

$$= \frac{(-r)(-r-1)\dots(-x-r+1)}{x!}\left(\frac{1}{p}\right)^{-r}(-q)^x$$
$$= \binom{-r}{x}\left(-\frac{q}{p}\right)^x\left(\frac{1}{p}\right)^{-r-x}$$
$$= \binom{-r}{x}P^xQ^{-r-x}, \quad \text{where } P = -\frac{q}{p}, Q = \frac{1}{p}$$

which is in the form of Binomial Distribution.

$$E(X) = np = (-r)\left(-\frac{q}{p}\right) = \frac{rq}{p}$$
$$V(X) = npq = (-r)\left(-\frac{q}{p}\right)\left(\frac{1}{p}\right) = \frac{rq}{p^2}$$

5 Probability Model

Consider a sequence of independent Bernoulli trails with probability of success p. Let X be the no of failure before the rth success. So, $P(X = x)$
$= P$ [there are $(r-1)$ success in the first $(x-r+1)$ trails and the $(x+r)$th trail results in a success]
$= P$ [$(r-1)$ success in $(x+r-1)$ trails] P(success)
$= \binom{x+r-1}{x}p^{r-1}q^xp$
$= \binom{x+r-1}{x}p^rq^x, x = 0, 1, 2, \dots$
This is the PMF of NB(r, p) distribution.
Let, Y be the number of trail needed to get the rth success. $Y = X + r$ with probability 1 .

$$P(Y = y) = P(X = y - r) = \binom{y-r+r-1}{y-r}p^rq^{y-r}, y = r, r+1, \dots$$
$$E(Y) = E(X) + r = \frac{rq}{p} + r = \frac{r}{p}$$
$$V(Y) = V(X) = \frac{rq}{p^2} \quad [\because \text{ no change due to linear shift }]$$

6 Cumulative Distribution Function

$$P(X \leqslant k) = \sum_{x=0}^{k}\binom{x+r-1}{x}p^rq^x = g(p)(say)$$

Differentiating wrt p,

$$g'(p) = \sum_{x=0}^{k}\binom{x+r-1}{x}rp^{r-1}(1-p)^x - \sum_{x=0}^{k}\binom{x+r-1}{x}p^rx(1-p)^{x-1}$$
$$= \sum_{x=0}^{k}\binom{x+r-1}{x}rp^{r-1}(1-p)^x - \sum_{x=0}^{k}\frac{(x+r-1)!}{(x-1)!(r-1)!}p^r(1-p)^{x-1}$$
$$= \sum_{x=0}^{k}\binom{x+r-1}{x}rp^{r-1}(1-p)^x - r\sum_{x=1}^{k}\binom{x+r-1}{x-1}p^r(1-p)^{x-1}$$
$$= \sum_{x=0}^{k}\left\{\binom{x+r-1}{x} + \binom{x+r-1}{x-1}\right\}rp^{r-1}(1-p)^x - r\sum_{x=1}^{k}\binom{x+p-1}{x-1}(1-p)^{x-1}p^{r-1}$$
$$= \sum_{x=0}^{k}\binom{x+r}{x}rp^{r-1}(1-p)^x - r\sum_{x=1}^{k}(x+r-1)(1-p)^{x-1}p^{r-1}$$

$$= \sum_{x=0}^{k} \binom{x+r}{x} r p^{r-1}(1-p)^x - r \sum_{x-1}^{k-1} \binom{x'+r}{x'} (1-p)^{x'} p^{r-1}$$

$$= \binom{k+r}{k} r p^{r-1}(1-p)^k [= g(p)]$$

$$g'(p) = \binom{r+k}{k} r p^{r-1}(1-p)^k$$

$$g'(t) = \binom{r+k}{k} r t^{r-1}(1-t)^k$$

$$\int_0^r g'(t)dt = \int_0^p \binom{r+k}{k} r t^{r-1}(1-t)^k dt$$

$$\Rightarrow g(p) - g(0) = \frac{\int_0^p t^{r-1}(1-t)^k dt}{B(r, k+1)}$$

$$\Rightarrow g(p) - 0 = \sum_{x=0}^{k} \binom{x+r-1}{x} p^r q^x = g(p) = I_p(r, k+1)$$

It is an increasing function of p and decreasing function of q.

7 Mode

$$f(x) = \begin{cases} \binom{x+r-1}{x} p^r q^x, & x = 0, 1, \ldots, p+q=1 \\ 0 & \text{ow} \end{cases}$$

$$\frac{f(x)}{f(x-1)} = \frac{\binom{x+r-1}{x} p^r q^x}{\binom{x+r-2}{x-1} p^r q^{x-1}} = q \frac{x+r-1}{x}$$

This is according as,

$$\frac{x+r-1}{x} q \gtreqless 1$$

$$\Rightarrow (x+r-1)q \gtreqless x$$

$$\Rightarrow x(q-1) \gtreqless (1-r)q$$

$$\Rightarrow x \gtreqless \frac{(r-1)q}{p}$$

Case I: $\frac{(r-1)q}{p}$ is not an integer, let $k = \left[\frac{(r-1)q}{p}\right]$ (say),

$$\frac{f(1)}{f(0)} > 1, \frac{f(2)}{f(1)} > 1, \frac{f(3)}{f(2)} > 1, \ldots, \frac{f(k)}{f(k-1)} > 1, \frac{f(k+1)}{f(k)} < 1, \ldots$$

$$\therefore \quad f(0) < f(1) < \ldots < f(k-1) < f(k) > f(k+1) > f(k+2) > \ldots$$

$\Rightarrow f(k)$ is the maximum and $k = \left[\frac{(p-1)q}{p}\right]$ is the mode.

Case II: $\frac{(r-1)q}{p}$ is an integer, (say k).

$$f(0) < f(1) < f(2) < \ldots < f(k-1) = f(k) > f(k+1) > f(k+2) > \cdots$$

$\therefore f(k)$ or $f(k-1)$ would be the maximum. $k = \frac{(r-1)q}{p}$ is the mode.

8 Tail Function

$$P(X > k) = 1 - P(X \leqslant k) = 1 - I_p(r, k+1)$$

$$
\begin{aligned}
&= 1 - \frac{\int_0^p t^{r-1}(1-t)^{\overline{k+1}-1}dt}{B(r,k+1)} \\
&= \frac{\int_0^1 t^{r-1}(1-t)^{\overline{k+1}-1}dt - \int_0^p t^{r-1}(1-t)^{\overline{k+1}-1}dt}{\int_0^1 t^{r-1}(1-t)^{\overline{k+1}-1}dt} \\
&= \frac{\int_p^1 (1-t)^{\overline{k+1}-1}t^{r-1}dt}{B(r,k+1)} \\
&= \frac{\int_q^0 -z^{\overline{k+1}-1}(1-z)^{r-1}dz}{B(r,k+1)} \\
&= \frac{\int_0^q z^{\overline{k+1}-1}(1-z)^{r-1}dz}{B(r,k+1)} \\
&= I_q(k+1,r)
\end{aligned}
$$

3. Let X be the intensity of an accident proneness of an individual and let Y be the no of accident of the individual. Let,

$$
f_X(x) = \begin{cases} \frac{\theta^\alpha}{\Gamma\alpha} e^{-\theta x} x^{\alpha-1}, & x \geqslant 1, \theta > 0, \alpha > 0 \text{ (real no)} \\ 0, & \text{otherwise} \end{cases}
$$

$$
f_{Y|X}(y \mid x) = \begin{cases} e^{-x}\frac{x^y}{y!}, y = 0,1,2,\ldots \\ 0, \text{ ow} \end{cases}
$$

$$
\begin{aligned}
f_Y(y) &= \int_{-\infty}^{\infty} f(y \mid x) f(x) dx, \quad y \in \mathbb{R}. \\
&= \int_0^\infty e^{-x}\frac{x^y}{y!}\frac{\theta^\alpha}{(\alpha-1)!}e^{-\theta x}x^{\alpha-1}dx \\
&= \frac{\theta^\alpha}{y!(\alpha-1)!}\int_0^\infty e^{-x(1+\theta)}x^{y+\alpha-1}dx \\
&= \frac{\theta^\alpha}{y!(\alpha-1)!}\frac{(\alpha+y-1)!}{(1+\theta)^{\alpha+y}} \\
&= \binom{y+\alpha-1}{y}\left(\frac{\theta}{1+\theta}\right)^\alpha\left(1-\frac{\theta}{1+\theta}\right)^y, y = 0,1,2,\ldots \\
Y &\sim \text{NB}\left(\alpha, \frac{\theta}{1+\theta}\right)
\end{aligned}
$$

4. A certain mathematician always carries one match box in his right pocket and one in his left. When he wants a match he selects a pocket at random and choose a match. Suppose, initially each box contained exactly N matches and consider the moment when for the first time our mathematician discovers that one box is empty. At that point in time, find the probability that the other box contain exactly r matches. Find E(X).

Let us identify the success with the choice of the left pocket. The left pocket box will be empty when the right pocket box contains r matches iff exactly ($n-r$) failure preceding (N + r) of success. A similar argument applies to the right pocket box.

$$
\begin{aligned}
p_r &= 2\binom{N-r+\overline{N+1}-1}{N-r}\left(\frac{1}{2}\right)^{N+1}\left(\frac{1}{2}\right)^{N-r} \quad [\text{ left } + \text{ right }] \\
&= \binom{2N-r}{N-r}\left(\frac{1}{2}\right)^{2N-r}, r = 0,1,\ldots N
\end{aligned}
$$

Clearly, p_r forms a PMF of a probability distribution as $\sum_{r=0}^N p_r = 1$.

$$
\begin{aligned}
\mu = \sum_{r=0}^{N} r p_r &= \sum_{r=0}^{N} r \binom{2N-r}{N-r} \left(\frac{1}{2}\right)^{2N-r} \\
(N-\mu) &= \sum_{r=0}^{N} (N-r) p_r \\
&= \sum_{r=0}^{N} (N-r) \frac{(2N-r)!}{(N-r)!N!} \left(\frac{1}{2}\right)^{2N-r} \\
&= \sum_{r=0}^{N-1} \frac{(2N-r)}{2} \frac{(2N-r-1)!}{(N-r-1)!n!} \left(\frac{1}{2}\right)^{2N-r-1} \\
&= \sum_{r=0}^{N-1} \frac{2N-r}{2} \binom{2N-r-1}{N-r-1} \frac{1}{2^{2N-r-1}} \\
&= \sum_{r=0}^{N-1} \frac{(2N+1)-(r+1)}{2} p_{r+1} \\
&= \frac{2N+1}{2} \sum_{r=0}^{N-1} p_{r+1} - \frac{1}{2} \sum_{r=0}^{N-1} (r+1) p_{r+1} \\
&= \frac{2N+1}{2} \sum_{r+1=k=1}^{N} p_k - \frac{1}{2} \sum_{r=0}^{N-1} (r+1) p_{r+1} \\
\Rightarrow (N-\mu) &= \frac{2N+1}{2} (1-p_0) - \frac{\mu}{2} \\
\Rightarrow \quad -\frac{\mu}{2} &= \frac{2N+1}{2} \left\{ 1 - \binom{2N}{N} \left(\frac{1}{2}\right)^{2N} \right\} - N \\
\Rightarrow \quad \frac{\mu}{2} &= N - \frac{2N+1}{2} \left\{ 1 - \binom{2N}{N} \left(\frac{1}{2}\right)^{2N} \right\} \\
&= N - \frac{2N+1}{2} + \frac{2N+1}{2} \binom{2N}{N} \left(\frac{1}{2}\right)^{2N} \\
&= \frac{2N-2N-1}{2} + \frac{2N+1}{2} \binom{2N}{N} \left(\frac{1}{2}\right)^{2N} \\
\Rightarrow \quad \mu &= (2N+1) \binom{2N}{N} \left(\frac{1}{2}\right)^{2N} - 1
\end{aligned}
$$

9 Relationship with Binomial Distribution

If $X \sim \mathrm{NB}(r,p)$,

$$
\begin{aligned}
P(X \leqslant k) &= P(U \geqslant r) \quad , U \sim \mathrm{Bin}(k+r, p) \\
&= P(V \leqslant k) \quad , V \sim \mathrm{Bin}(k+1, q)
\end{aligned}
$$

X denotes the number of failure preceding the rth success and $(a+X)$ denotes the no of trails required to get the rth success.

$$
\begin{aligned}
P(X \leqslant k) &= P(X + r \leqslant k + r) \\
&= P(\text{ no of trails required to get rth success } \leqslant r+k)
\end{aligned}
$$

If we define U as the no of success in $(r+k)$ trails,

$$
\begin{aligned}
&= P(U \geqslant r) && \text{So, } U \sim \text{Bin}(k+r, p)\\
&= P(-U \leqslant -r)\\
&= P(k+r-U \leqslant k) && (K+r-U) \sim \text{Bin}(k+r, q)\\
&= P(V \leqslant k) && \Rightarrow V \sim \text{Bin}(k+r, q)
\end{aligned}
$$

5. Let X be a random variable with PMF,

$$
f(x) = \begin{cases} c\binom{2n-x}{n}2^x & , x = 0, 1, \ldots n \\ 0 & , \text{ ow} \end{cases}
$$

Determine c such that it would be a PMF.

$$
\begin{aligned}
& \sum_{x=0}^{n} f(x) = 1\\
\Rightarrow \quad & c\sum_{x=0}^{n} \binom{2n-x}{n} 2^x = 1\\
\Rightarrow \quad & c\sum_{y=0}^{n} \binom{n+y}{n} 2^{n-y} = 1, \quad n-x = y\\
\Rightarrow \quad & c2^{2n+1} \sum_{y=0}^{n} \binom{y+(n+1)-1}{y} \left(\frac{1}{2}\right)^{n+1} \left(\frac{1}{2}\right)^{y} = 1
\end{aligned}
$$

Now,

$$
\begin{aligned}
& P(Y \leqslant n), Y \sim \text{NB}\left(n+1, \frac{1}{2}\right)\\
& P(V \leqslant n), V \sim \text{Bin}\left(2n+1, \frac{1}{2}\right)\\
\Rightarrow \quad & \sum_{V=0}^{n} \binom{2n+1}{V} \left(\frac{1}{2}\right)^{2n+1} = \frac{1}{2}\\
& \Rightarrow c2^{2n+1}\frac{1}{2} = 1\\
& \Rightarrow c = 2^{-2n}\\
f(x) = & 2^{-2n}\binom{2n-x}{n}2^x = \binom{2n-x}{n}\left(\frac{1}{2}\right)^{2n-x}
\end{aligned}
$$

Alternative, $c2^{2n+1} \underbrace{P(Y \leqslant n)}_{I_{1/2}(n+1,n+1)} = 1, \quad Y \sim NB\left(n+1, \frac{1}{2}\right)$

$$
\begin{aligned}
& \Rightarrow c2^{2n+1} I_{1/2}(n+1, n+1) = 1\\
& \Rightarrow c2^{2n+1} \frac{\int_0^{1/2} t^n(1-t)^n dt}{B(n+1, n+1)} = 1\\
& \Rightarrow c2^{2n+1} \frac{\int_0^{1/2} t^n(1-t)^n dt}{2\int_0^{1/2} t^n(1-t)^n dt} = 1\\
& \Rightarrow c2^{2n+1}\frac{1}{2} = 1\\
& \Rightarrow c = \left(\frac{1}{2}\right)^{2n}
\end{aligned}
$$

Putting, the value of c, we will proof it is a pmf.

$$
\begin{aligned}
&\sum_x f(x)\\
=&2^{-2n}\sum_{x=0}^{n}\binom{2n-x}{n}2^x, \quad n-x=y\\
=&2^{-2n}\sum_{y=0}^{n}\binom{n+y}{y}2^{n-y}\\
=&2^{-2n}2^{2n+1}\sum_{y=0}^{n}\binom{y+(n+1)-1}{y}\left(\frac{1}{2}\right)^{n+1}\left(\frac{1}{2}\right)y\\
=&2P(Y\leqslant n)\\
=&2I_{1/2}(n+1,n+1)\\
=&2\frac{1}{2}\\
=&1.
\end{aligned}
$$

6. In a series of indefinite Bernoulli trails with success probability p. Find the probability that a successes will appear before b failures (Pascal format).

Let X denote the no of failures preceding ath success.

$$
\begin{aligned}
x &\sim \mathrm{NB}(a,p)\\
P(X\leqslant b-1)&=\sum_{x=0}^{b-1}\binom{x+a-1}{x}p^aq^x
\end{aligned}
$$

Alternative,

Y denotes the no of failures in $(a+b-1)$ trails.

$$
P(Y\leqslant b-1)=\sum_{y=0}^{b-1}\binom{a+b-1}{y}q^yp^{a+b-1-y}
$$

So, $\quad P(X\leqslant b-1)=P(Y\leqslant b-1)$

$$
\begin{aligned}
&\Rightarrow\sum_{i=0}^{b-1}\binom{a+i-1}{a-i}p^\alpha q^i=\sum_{i=0}^{b-1}\binom{a+b-1}{i}q^ip^{a+b-1-i}\\
&\Rightarrow\sum_{i=0}^{b-1}\binom{a+i-1}{a-i}=\sum_{i=0}^{b-1}\binom{a+b-1}{i}p^{b-i-1}
\end{aligned}
$$

7. Prove that, if $X_1\sim \mathrm{NB}(r_1,p), X_2\sim \mathrm{NB}(r_2,p)$, both are independent to each other, $X_1+X_2\sim \mathrm{NB}(r_1+r_2,p)$.

$$
\begin{aligned}
P(X_1+X_2=x)&=\sum_x P(X_1=x)\,P(X_2=y-x)\\
&=\sum_x\binom{x+r_1-1}{r_1-1}p^{r_1}q^x\binom{y-x+r_2-1}{r_2-1}p^{r_2}q^{y-x}\\
&=\sum_x\binom{x+r_1-1+y-x+r_2-1}{r_1+r_2-2}p^{r_1+r_2}q^y\\
&=\sum_y\binom{y+r_1+r_2-2+1}{r_1+r_2-2}p^{r_1+r_2}q^y\\
&=\sum_y\binom{y+r_1+r_2-1}{y}p^{r_1+r_2}q^y
\end{aligned}
$$

$$
\therefore Y\sim NB(r_1+r_2,p)
$$

8. $X_i \sim \mathrm{NB}\,(r_i, p)$ i = 1,2. Find the conditional distribution of $(X_1 \mid X_1 + X_2 = y)$

$$\begin{aligned} P\,(X_1 = x \mid X_1 + X_2 = y) &= \frac{P\,(X_1 = x)\,P\,(X_2 = y - x)}{P\,(X_1 + X_2 = y)} \\ &= \frac{\binom{x+r_1-1}{x} p^{r_1} q^x \binom{y-x+r_2-1}{y-x} p^{r_2} q^{y-x}}{\binom{y+r_1+r_2-1}{y} p^{r_1+r_2} q^y} \\ &= \frac{\binom{x+r_1-1}{r_1-1}\binom{y-x+r_2-1}{r_2-1}}{\binom{y+r_1+r_2-1}{r_1+r_2-1}} \end{aligned}$$

It is independent of parameters.

Remark:

In $\mathrm{Bin}(n, p)$ distribution, we take a fixed number of trail, then we count no of success but in NB(r, p) distribution we fixed up the no of success and then we count the no of trail required to get the rth success. Hence, the NB distribution (Pascal's) is also known as Inverse Binomial Sampling.

9. $X \sim \mathrm{Bin}(n, p)$, show that $E\left(\frac{1}{1+X}\right) = \frac{p}{q(r-1)}\left\{1 - p^{r-1}\right\}$

$$\begin{aligned} E\left(\frac{1}{1+x}\right) &= \sum_{x=0}^{\infty} \frac{1}{1+x}\binom{x+r-1}{x} p^r q^x \\ &= \sum_{x=0}^{\infty} \frac{(x+r-1)!}{(x+1)!(r-1)!} p^r q^x \\ &= \sum_{x=0}^{\infty} \frac{(\overline{x+1} + \overline{r-1} - 1)!}{(x+1)!(r-1)!} p^r q^x \\ &= \frac{p}{q(r-1)} \sum_{x=0}^{\infty} \frac{(\overline{x+1} + \overline{r-1} - 1)!}{(x+1)!(r-2)!} p^{p-1} q^{x+1} \\ &= \frac{p}{q(r-1)} \sum_{x+1}^{\infty} y = 1 \\ &= \frac{p}{q(r-1)}[1 - p(X=0)] \\ &= \frac{p}{q(r-1)}(1 - p^{r-1})! \end{aligned}$$

10 Mean Deviation about mean

$$\begin{aligned} \mathrm{MD}_\mu(x) &= E[|X - \mu|] \\ &= \sum_{x=0}^{\infty} |x - \mu| f(x) \\ &= \sum_{x=0}^{[\mu]} (\mu - x) f(x) + \sum_{x=[\mu+1]}^{\infty} (x - \mu) f(x) \\ &= 2 \sum_{x=0}^{[\mu]} (\mu - x) f(x) \\ &= 2 \sum_{x=0}^{[\mu]} \left(\frac{rq}{p} - x\right) f(x) \end{aligned}$$

$$
\begin{aligned}
&= \sum_{x=0}^{[\mu]}(\mu - x)f(x) + \sum_{x=[\mu+1]}^{\infty}(x-\mu)f(x) \\
&= 2\sum_{x=0}^{[\mu]}(\mu - x)f(x) \\
&= 2\sum_{x=0}^{[\mu]}\left(\frac{rq}{p} - x\right)f(x) \\
&= \frac{2}{p}\sum_{x=0}^{[\mu]}(rq - px)f(x) \\
&= \frac{2}{p}\sum_{x=0}^{[\mu]}[(r+x)qf(x) - xf(x)]
\end{aligned}
$$

Alternative,

$$
\begin{aligned}
g(x) &= xf(x) \\
g(x+1) &= (x+1)f(x) \\
&= (x+1)\binom{x+1+r-1}{x+1}p^r q^{x+1} \\
&= (x+1)\frac{(x+r)!}{(x+1)!(r-1)!}p^r q^{x+1} \\
&= (r+x)q\frac{(x+r-1)!}{x!(r-1)!}p^r q^x \\
&= (x+r)qf(x)
\end{aligned}
$$

$$
\begin{aligned}
\mathrm{MD}_\mu(x) &= \frac{2}{p}\sum_{x=0}^{[u]}\{(x+r)qf(x) - xf(x)\} \\
&= \frac{2}{p}\sum_{x=0}^{[u]}\{g(x+1) - g(x)\} \\
&= \frac{2}{p}\sum_{x=[u]+1}^{\infty}\{g(x+1) - g(x)\} \\
&= \frac{2}{p}\{1 - g[u]\}
\end{aligned}
$$